What topology hold-out removes—and what it does not

Running title: What does topology hold-out actually test?

Tao Wang[1], [1] College of Chemical Engineering, Nanjing Tech University, Nanjing 211816, China *Correspondence: <3598260175@qq.com>*

Abstract

Machine-learning screens of metal–organic frameworks (MOFs) for CO_2 capture increasingly report the validation protocol that produced a score, and topology hold-out has become the standard stricter alternative to random splitting. Prior work already reads topology hold-out as a combined structural and chemical distribution shift; this work measures the size of the chemistry that the split withholds. On the hypothetical-MOF library hMOF (324,426 structures), 74.7% of coarse-grained chemical compositions occur in a single topology, accounting for 61.3% of structures, and seven of eleven topologies leave every structure without a chemical counterpart in training. On the one usable chemistry-matched pair (pcu ↔ nbo), removing the partner topology raised nbo-side MAE by 70.0% while a size-matched control did not, and removing only the shared-composition partner rows improved pcu-side MAE, so partner coverage is direction-dependent rather than uniformly beneficial. A chemistry-key hold-out produces only 21.1% of the topology hold-out's R^2 decrease, so the support loss is real but does not translate proportionally into performance loss. Across five independently redrawn 25,000-sample support sets, the top-1% hit rate fell from 50.4% to 19.3% under topology-grouped validation. A support audit in ARC-MOF, which is partly dependent on hMOF and uses different composition keys, confirms strong coupling under bounded sensitivity rather than as an independent replication. We retain topology hold-out as a stress test while recommending that grouped-split reports disclose their chemical support, deployment-relevant aggregation and ranking performance.

Keywords: metal–organic frameworks; CO_2 capture; machine learning; data leakage; confounding; topology hold-out; grouped cross-validation

1. Introduction

1.1 The protocol that decides the claim

Machine-learned screens of MOFs report high random-split R^2 for CO_2 uptake — 0.93–0.97 at 0.15–2.5 bar in ref. 17 — and for other MOF properties (ref. 26), and such numbers are used in materials-selection workflows for post-combustion CO_2 capture at 0.15 bar and 298 K (refs. 2, 4); under a topology hold-out the same class of model spans R^2 below 0.10 to ≥ 0.97 depending on the target and the pressure range (ref. 8). The dominant hMOF database is **enumerated**: a finite set of building blocks is recombined on a library of topological templates (ref. 1), so many entries are close relatives and a random split places relatives on both sides of the train/test boundary. ARC-MOF is a mixed-source library (refs. 3, 15) that partly overlaps hMOF (§3.5). The result is a familiar but under-specified situation: the number a screen reports depends on who ends up on which side of the boundary, and the boundary is set by the protocol, not by the data.

Topology hold-out evaluates prediction on unseen topology groups. When chemistry is strongly coupled to topology, that task also entails chemical-support shift; its score alone cannot isolate an effect attributable to topology. This

work quantifies the support the split removes and examines the consequences through controlled training-pool contrasts and screening metrics.

1.2 What prior work establishes, and what this work adds

The shift from reporting a score to reporting the protocol that produced it is recent and well documented. Three 2026 studies are the clearest points of reference, and each overlaps this work in a specific way. Asgari et al. held the model, descriptors, hyper-parameters and metric fixed and varied only the split, finding that the split itself defines the discovery claim — under their design R^2 falls from 0.758 under random splitting to 0.267 under topology grouping, and top-5% recovery falls from 62.2% to 37.6% (ref. 16). Alimardani et al. extended that result across a common cohort of 255,939 MOFs and 27 adsorption targets under five descriptor families, showing that descriptor adequacy is regime-dependent — at 5.8 bar, geometry-only R^2 is ≈ 0.30 against ≈ 0.69 once chemistry is added (ref. 8). Wang et al., on the same database and the same 0.15 bar target used here, found by ablation that chemistry-type descriptors contribute most of the gain beyond geometry (ref. 17).

These studies are careful about what they read a topology hold-out as. Alimardani et al. interpret it "as a deployment proxy for structural extrapolation" while stressing that it "does not imply that topology labels are the mechanistic cause of prediction error" (ref. 8). Asgari et al. read topology- and node-family hold-outs as regimes that "combine chemical-family transfer with descriptor-space shift", describing the topology-grouped penalty as "a combined stress test involving topology-label granularity, group-size imbalance, descriptor coverage, descriptor-space shift, and target sensitivity" rather than a statement that topology alone mechanistically determines adsorption (ref. 16). Related analyses by the same research group go further in the same direction: they build chemically auditable descriptor atlases (ref. 9), treat benchmark maturity as a precondition for a reliable conclusion (ref. 10), and ask when topology matters beyond pore metrics (ref. 28). Prior work therefore already treats the protocol as a mixed stress test rather than a clean separation of topology from chemistry, and this paper claims no priority on either finding.

This work measures the size of the mixing. The three studies compared above do not report the composition-to-topology determinism — whether as the single-topology composition fraction or as H(topology | chemistry) — nor do they connect such a number to the support a topology hold-out actually removes from training. Related work shows topology is very unevenly distributed across MOF databases (ref. 7), that MOF data carry known biases and redundancy (ref. 6), and that identifier schemes such as MOFkey exist precisely to detect structural duplicates (ref. 27); these are descriptions of the data, not of the hold-out. A grouped split remains a legitimate deployment proxy even when the blocking variable is confounded with the predictors; the prior-art point (refs. 18–20) is only that the confound changes what the score can be read as, which is why the chemical support must be disclosed alongside the score. The same tension is documented outside MOFs: scaffold and time splits are standard in molecular benchmarks because random splits overstate prospective accuracy (refs. 21–23), and the graph-based model classes now common in this literature (ref. 24) are trained and evaluated under the same split definitions, so the choice of split bears on them as well. Molecular out-of-distribution benchmarks, QSAR partitioning studies and chronological evaluations all make the split itself the variable under test (refs. 29–31). What this work adds is the measured magnitude of the coupling and its consequence, in hMOF (an enumerated library) and in ARC-MOF (a mixed-source library); whether the same magnitude holds for enumerated libraries generally is not tested here. Crucially, a measured support loss is not a measured error attribution: the counts fix the identity of the withheld chemistry, not the cost of withholding it.

1.3 Contributions

Three contributions follow. (i) A support audit — the single-topology composition fraction, the conditional-entropy ratio and its reverse, and the pairwise-overlap profile — that quantifies how much chemistry a grouped split withholds, supplied as a portable diagnostic for arbitrary CSV input. (ii) A controlled intervention on the one usable chemistry-matched pair, with size-matched, composition-matched and shared-composition arms, showing that removing the partner raises nbo-side MAE by 70.0% while removing only the shared-composition rows improves pcu-side MAE — a direction dependence not explained by random thinning to the same size. (iii) A screening-level consequence and a 22-rule reporting checklist (kept in full in the Supporting Information and its companion tool) that let any grouped-split report disclose its own confound structure before it reports a score.

---

2. Methods

2.1 Data, sampling and composition definitions

The primary database is the hMOF release (ref. 2) — the enumerated library built with the procedure of ref. 1 — comprising 324,426 structures; ref. 1 is cited for the enumeration method and ref. 2 for the release, and the archived record under Data availability is the file actually used. The hMOF release contains 42 fields; models use eight numerical structure descriptors and the categorical feature blocks specified in §2.3 (the SI parameter table lists the field groups). The elite release subset is 8,206 structures. The second database is ARC-MOF v6 (ref. 15), 264,225 structures, from which only the topology and chemical-cluster metadata are used, because it carries no adsorption property in a form this study's models consume, so it cannot serve as an external prediction test. Both are public; neither is redistributed (Data availability). Modelling samples of 25,000, 50,000 or 100,000 rows are drawn with a fixed seed unless a multi-seed design states otherwise: the 25,000-sample scale is the screening experiment of §3.4, the 50,000-sample scale is the controlled intervention of §3.3 (large enough that the pcu ↔ nbo shared compositions carry usable row counts, small enough to keep the many-arm refit cheap), and the 100,000-sample scale is the protocol comparison of §3.2. The full-database support figures use all rows and are always reported with their n, because the single-topology fraction is sample-size dependent: it reads 98.9% at n = 25,000 against 93.2% at n = 324,426 under the fine definition, so a number without n is not comparable to anything.

Chemical composition has two granularities. The **coarse** definition — the headline — concatenates the metal centre and the two organic linkers (the `metal_linker`, `organic_linker1` and `organic_linker2` fields); the **fine** definition adds the functional group (`functional_groups`). A structure whose key has no exact counterpart in training is **unseen** under that definition. The two databases define "a composition" with different keys: hMOF uses building-block identities, whereas ARC-MOF uses its own metal/linker/functional cluster identifiers, so the two are not interchangeable and cross-database magnitudes are never ranked against each other. For hMOF, the 1,997 structures with an unannotated functional group are treated as one additional fine-grained value rather than dropped, so that the fine definition does not silently discard rows.

2.2 Support, entropy and overlap measures

Three quantities describe the support a hold-out removes. (i) The **single-topology fraction** — the share of distinct compositions that occur in exactly one topology — weights compositions equally and answers "how much of the chemistry space is private to one topology?". (ii) The **structure-level unseen fraction** — the share of structures whose key has no training counterpart after their topology is held out — weights structures and answers "how much of the database loses its same-key counterpart?". The two differ whenever common compositions are more populous,

so both are always reported. (iii) The **conditional-entropy ratio** H(topology | chemistry) / H(topology) is structure-weighted because the deployment question is about structures, not composition labels: a composition that appears 10,000 times should weigh 10,000 times more in the entropy than one that appears once, and the structure weighting makes the ratio answer "how much of the structure mass does conditioning on chemistry resolve?". The relation is asymmetric, so the reverse direction is also reported: H(chemistry | topology) / H(chemistry) = 0.860 means topology removes only 14% of composition uncertainty. Quoting 18.0% alone would invite the false reading that the two variables mutually determine each other.

The plug-in entropy carries a first-order Miller–Madow correction $\Delta H(T|C) = (K_{TC} - K_C)/(2\,N \ln 2)$, where K_C and K_{TC} count occupied marginal and joint cells. For the sparse fine-granularity space this correction is substantial (≈ 22% of the estimate, 0.0307 bit against $H(T|C) = 0.139$ bit) and is reported as a sensitivity check rather than a proof of unbiasedness, because a space with many singletons need not have its bias removed by a first-order term; the coarse correction is negligible (0.000912 bit). The pairwise-overlap profile completes the audit: it lists, for every topology pair, the count of compositions they share and the Jaccard index (the shared count over the union of the two composition sets), which is the quantity that determines whether a controlled contrast is possible at all — a pair with a Jaccard of 0.000 has no shared chemistry and can serve no matched-pair experiment.

2.3 Models, features and validation protocols

The working model is a random forest (120 estimators by default; three designs use 100 — the ten-target panel of SI-G, the five-seed 25,000-sample LOTO runs of SI-D, and the S_g curve of SI-B — all recorded in the SI parameter table), held fixed within every comparison so that no comparison confounds a protocol change with a model change. A random forest is used because it is fast to refit over the many folds and arms, it handles the one-hot identity columns without tuning, and its decisions are attributable, which the SHAP diagnostics of SI-E use (ref. 25); the choice is a convenience, not a claim that it is the best model. The default feature set is 426 columns: eight structure-derived descriptors (volume, weight, surface area, void fraction, void volume and three pore-diameter descriptors) plus building-block one-hot blocks for topology (11 levels), metal centre (7) and functional groups (400), i.e. 8 + 11 + 7 + 400 = 426. The linker pair (organic_linker1 + organic_linker2) is **not** a modelling column: it enters only as a grouping variable (§3.2, §4.2) and in the omitted-variable analysis, never as a feature. A 415-column variant removes the topology block from both arms of a split comparison, so the two arms stay matched on the number of identity columns; the topology block is the focus of the check because it is the feature that differs structurally between the two split arms, and removing it from both is the only feature-level control the comparison needs. Comparisons that cross model families are not used for absolute levels: gradient boosting and ridge (refs. 32, 33; implemented with refs. 11–14) are refit only to check that the topology penalty is not specific to one family, and are single fits without tuning or seed intervals, so only within-family gaps are interpreted and no claim of model equivalence is made.

Two grouped-validation constructions appear and must not be conflated. A **5-fold GroupKFold** by topology (the §3.2 baseline) gives a pooled R^2 of 0.3786. A **leave-one-topology-out** (LOTO) scheme, scored once per held-out topology, gives the four aggregation rules compared in §3.2. The paired protocol comparisons of this study are designed so that two arms share the same test target values and total sum of squares; their prediction-error sums of squares then differ, which is exactly the quantity the comparison measures — a shared denominator is not a shared numerator. R^2 across different test populations can still be reported, but such a difference cannot be attributed solely to the model or protocol, because the denominator differs too.

2.4 Fixed-test-set interventions (arms a–f)

The controlled contrast uses 50,000 samples and the single chemistry-matched pair pcu ↔ nbo (408 shared compositions). For each of five independent seeds, the test set is drawn once and held fixed; only the training pool changes across arms: **(a)** with the partner topology included; **(b)** with the partner topology removed; **(c)** a size-matched random draw from (a) down to (b)'s training size; **(d)** a draw matching (b)'s composition/topology profile; **(e)** removing only those partner rows whose composition also occurs in the test set; **(f)** a size-matched random draw from (e). The seed-fixed test set is what makes R^2, MAE and RMSE directly comparable across arms, because the target variance of the test set does not change between arms — the only thing that changes is which training rows the model sees, which is the variable the experiment manipulates. Each arm answers the specific question the previous one leaves open, which is why they are presented in the fixed order a–f. Two size-matched (c) designs exist and are kept separate (SI-C.1): a **nested 5 × 5 draw** (`diversity_control.py`, `outputs/composition_matched_control.csv`), whose statistic is the seed-level mean of that seed's five inner draws with the SD taken across the five outer seed means, and a **seed-42 repeated-draw** (`confound_diagnosis.py --size-match`, `outputs/isoreticular_control_sizematched.csv`); they are different replication designs, not two precisions of one estimate, and are never pooled into one error bar. Arm (d) reproduces (b)'s composition/topology profile, which confines it to (b)'s own row set, so it is an internal-consistency check rather than an independent mechanism contrast. The primary metric is MAE; RMSE and the deployable training-mean baseline are reported alongside.

2.5 Aggregation, replication and uncertainty

Grouped-split R^2 is reported under four rules: **pooled** (all fold predictions concatenated and scored once), **weighted** (per-fold R^2 weighted by test-set size), **macro** (unweighted mean of per-fold R^2) and **median**. All four are reported because no single rule is right for every deployment question — a whole-database deployment dominated by the large topologies is best described by pooled, a "typical per-topology" performance by macro or median — and they can differ in sign, so reporting one alone would hide the sign reversal. A fold whose test-set variance is zero makes R^2 undefined and is handled explicitly rather than silently dropped. Two distinct uncertainty objects are kept separate throughout: the **standard deviation over seeds** describes how a quantity varies across independent redraws, while the **paired 95% interval** describes the confidence on a paired difference over the same seeds, and neither is rewritten as the other; a per-seed percentage spread is a third object and does not describe a paired effect. Percentages quoted for a paired change scale the absolute paired difference by the observed arm-(a) mean, with that denominator treated as fixed; they are not ratio intervals in which numerator and denominator are estimated jointly. Five seeds are five redraws within one database, not five independent databases or five independent topology populations, so these intervals describe seed-to-seed variation under that fixed database. Five-seed designs use seeds 42, 143, 244, 345, 446; five seeds are used rather than one because the screening and intervention quantities are seed-sensitive (the topology overlap standard deviation is 37% of its mean).

2.6 Screening metrics and ARC sensitivity

Screening utility is measured on a support set shared within each seed: every sample carries an out-of-fold prediction from every strategy, so top-k overlap and regret are computed on identical rows and cannot be confounded by a change of denominator. These metrics pool out-of-fold predictions on the same candidate set; they describe retrospective validation-protocol performance rather than the shortlist of a single deployed model, because the contrast still includes differences in training-set composition and fold sizes even though the scoring rows are shared. Ties are resolved by a deterministic rule rather than by "take the first k": rows are sorted by score descending with a stable sort on the archived row order and the first k are taken, so a tie at the boundary is broken by row position —

arbitrary but reproducible. On the true-uptake side no tie straddles the top-1%, top-5% or top-10% threshold in any of the five 25,000-row seeds (checked); ties among the predicted scores were not separately audited, so the rule is documented but its effect on the reported overlaps is not measured. The **oracle** is the true top-k of the same samples, i.e. the materials a budget would select if the true uptake were known. k = 1% (250 materials at 25,000 samples) is the headline budget, chosen as a representative screening size; k = 5% and 10% are in the Supporting Information (SI-F). For ARC-MOF, two distinct controls are used and are not interchangeable: **within each source, topology labels are permuted across records while composition keys and topology-label counts are held fixed** (1,000 permutations), testing whether the observed composition–topology association exceeds that permutation reference; separately, a **common-rate binomial parametric bootstrap** (20,000 draws) assesses between-source heterogeneity. The two controls address different null hypotheses and neither identifies a database-construction mechanism (SI-H).

---

3. Results

3.1 Chemical support removed by topology hold-out

On the full hMOF database (324,426 structures), **74.7% of coarse compositions occur in a single topology**, and those compositions account for **61.3% of structures**. The two numbers answer different questions and must not be interchanged. The first weights compositions equally: of the 1,622 coarse compositions (metal centre plus the two linkers), 1,212 occur in exactly one topology, so nearly three-quarters of the chemistry space is private to one topology. The second weights structures: those 1,212 single-topology compositions account for 61.3% of the 324,426 structures, so a majority of the database loses its same-key training counterpart when its topology is held out. They differ because single-topology compositions are on average less populous than multi-topology ones, and quoting only 74.7% invites the false reading that 74.7% of structures are unseen. Both units are reported throughout because the distinction is the difference between "most chemistries" and "most structures".

The loss is strongly heterogeneous across topologies (Figure 1). Seven of the eleven topologies have no other topology sharing any of their composite chemistry keys, so holding one out leaves every structure of that topology unseen — a **complete** chemistry hold-out, in the sense that the composite key is wholly absent from the remaining training set. The seven are sra, etb, acs, bcu, pts, fof and the, and together they hold only **21.3%** of structures (23,962 + 19,682 + 13,655 + 8,795 + 1,522 + 1,256 + 352 = 69,224 of 324,426); "7 of 11 are exact hold-outs" is therefore not "most structures are exact hold-outs", and the two facts must be stated together to avoid overstating how much of the database is affected. The remaining four topologies are only partially covered: pcu leaves 60.1% of its structures unseen, nbo 12.3%, tbo 89.0%, and rht (14 structures, sharing two compositions with tbo) none. The complete eleven-row decomposition and the metal/linker/conjunction breakdown are in SI-B; the metal/linker/conjunction split shows that neither the metal centre nor the linker pair alone reproduces the conjunction's exclusivity, but the headline remains the composite key because that is what the topology hold-out actually withholds. The point the main text keeps is that "completely unseen" holds only for the composite chemistry key, not for each building block separately — a topology can have no composite-key counterpart while its metal centre or a linker still recurs in training.

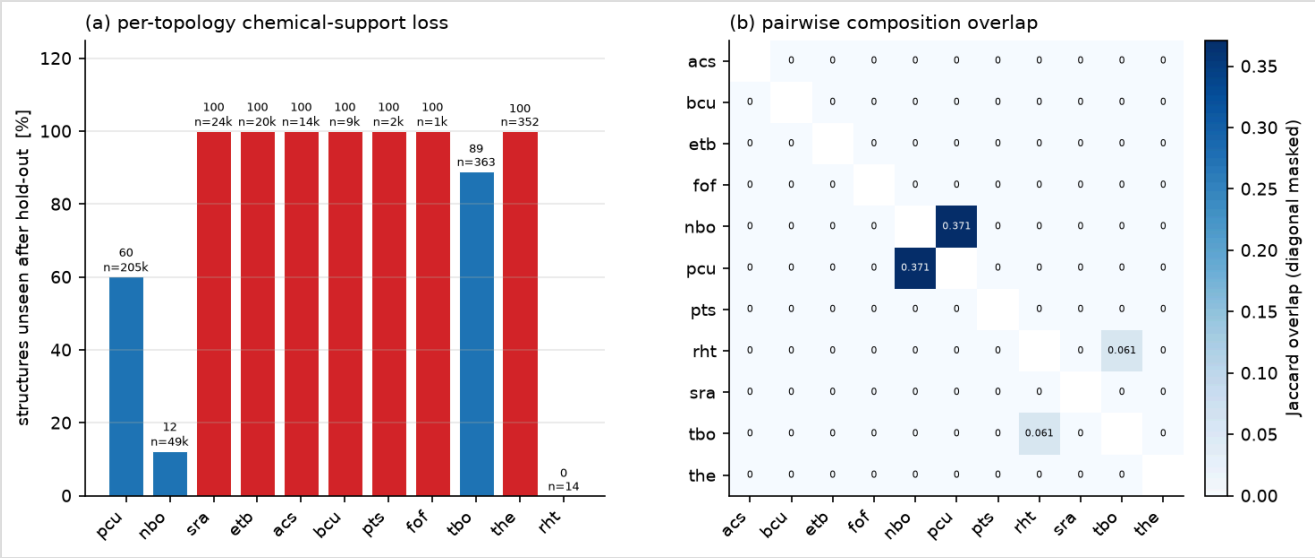


Figure 1. Chemical support removed by topology hold-out on hMOF. **(a)** Per-topology share of structures whose coarse composition key has no training counterpart after that topology is held out; each bar is labelled with that percentage and with the topology's structure count in thousands (n = 205k … 14), and seven of eleven topologies sit at 100% but hold only 21.3% of structures, while pcu (60.1%), nbo (12.3%) and tbo (89.0%) are partially covered and rht is fully covered. **(b)** Pairwise Jaccard overlap of the coarse composition sets; the diagonal is masked (self-overlap is not zero), and only pcu ↔ nbo (0.371) and rht ↔ tbo (0.061) are non-zero, leaving one usable contrast. *Source data:* `outputs/chemistry_seen_by_topology.csv` , `outputs/topology_pair_overlap.csv` .

Conditioning on composition removes **82.0%** of the entropy in the topology label (H(T|C)/H(T) = 0.180, structure-weighted), consistent with composition largely determining topology. The relation is asymmetric and both directions are reported: the reverse ratio H(C|T)/H(C) = 0.860 says topology leaves 86% of composition uncertainty, so 18.0% is the small direction of an asymmetric relation, not a symmetric measure of mutual determinism. The weighting matters and is stated once: the ratio is structure-weighted throughout, so the composition-weighted value (11.4%) differs from the headline and is normalised by the topology's structure-level entropy; quoting 18.0% alone would invite the false reading that the two variables mutually determine each other. The fine-granularity figure moves in the same direction (93.2% single-topology; 84.6% structures unseen), but for a different reason — requiring an exact functional-group match creates many singleton compositions that are single-topology by construction rather than by observation. Restricting to compositions that occur at least twice lowers the fine single-topology fraction to **79.1%** (66,172 non-singleton compositions), whereas the coarse headline is barely affected (only 21 of 1,622 coarse compositions are singletons; 74.4% among non-singletons against 74.7% overall). The fine-granularity robustness check is therefore doing exactly one job — confirming that the coupling's direction is not an artefact of the coarse key — while leaving its magnitude to the coarse headline, and the two values are not interchangeable. Fine-grained values are reported only as robustness checks with these caveats, not as headline evidence (SI-B).

For a practitioner, the practical reading is already fixed by these counts. A topology hold-out on this database is, in seven of eleven cases, an exact composite-key hold-out, so a score from such a fold is not "transfer to an unseen topology" but "transfer to an unseen topology-and-chemistry combination". In the remaining four cases it is a partial chemistry hold-out whose severity the per-topology unseen fraction quantifies. The practitioner who knows this can still use the protocol — it is a harder test than random splitting — but can no longer describe its result as a clean topology-transfer estimate.

The pairwise-overlap profile explains why the controlled experiment is possible at all. Of the 55 topology pairs, only pcu ↔ nbo (408 shared compositions) and rht ↔ tbo (2) share any composition; every other pair has Jaccard = 0.000. The rht ↔ tbo pair is unusable, so a single pair — pcu ↔ nbo — carries the entire controlled experiment, and the paper's causal-sounding claims are deliberately confined to that one pair. The 408 shared compositions are not a

bare label but a usable quantity of data: in the 50,000-sample protocol, deleting them in arm (e) removes 12,342 training rows on the nbo test side and 6,689 on the pcu test side, enough to move the fit detectably, which is what separates this pair from the two-composition rht ↔ tbo pair by two orders of magnitude. Two further facts bound how the headline is read. First, the fine-definition single-topology statistic is sample-size dependent — 98.9% at n = 25,000 against 93.2% at the full database, for the reason given in §2.1 — so the full-database number is the headline and every value carries its n. Second, a negative result closes the section: the per-topology unseen-chemistry rate does not detectably order the per-topology hold-out cost (Pearson r = +0.10, p = 0.78; Spearman ρ = +0.28, p = 0.44 across the ten topologies with n_test ≥ 100; SI-B). Support loss can therefore be reported as a structural fact, but it cannot be converted into a predicted error magnitude; the direction of any error ordering is not supported. This negative result is what forces the paper to earn its performance claims empirically in §3.2 and §3.3 rather than inferring them from the support counts.

3.2 Protocol contrasts distinguish support loss from performance loss

On 100,000 samples with the shared 426-feature set, the five validation schemes give pooled R^2 of 0.7529 (random), 0.6738 (chemistry key), 0.6606 (linker pair), 0.5270 (metal) and 0.3786 (topology, 5-fold GroupKFold) (Table 1). The observed order is chemistry key (0.674) > linker pair (0.661) > metal (0.527) > topology (0.379); this is a list of measured values, not a monotonic "coarsening" law, because the topology grouping withholds a structural family as well as its chemistry and is not simply a coarser member of the same chain. The chemistry-key hold-out therefore removes far less of the R^2 than the topology hold-out: holding out the composite chemistry key retains 0.674 against topology's 0.379. The two are not on identical footing as they stand — the chemistry arm keeps the 11 topology one-hot columns that the topology arm loses by construction, and the two protocols differ in fold structure and training size (the topology arm groups into 11 topologies over 5 folds, training on 36,553 to 92,965 records, while the chemistry arm groups into 1,590 composition keys over 5 folds, training on 80,000 each) — so the comparison was repeated with the topology block removed from **both** arms.

The chemistry-key decrease is **21.1%** of the topology-grouped decrease with 426 features and **21.5%** with matched 415-feature sets (matched decrease 0.3603). The 415-feature check exists to rule out one specific objection — that the chemistry arm is information-advantaged because it keeps the topology one-hot block that the topology arm loses by construction. Removing that block from both arms moves the chemistry arm by only −0.0065 R^2, so the ordering is not an artefact of the unmatched feature count. These are **descriptive ratios between validation protocols**, not causal shares of the penalty: the two protocols differ in fold structure, training-set size, chemical coverage and group structure, and matching 415 features does not match any of those. The complement of the ratio is deliberately not named as a mechanism — neither "chemistry" nor "geometry-to-uptake mapping" is claimed to account for the remaining ~79%, because the design does not identify it. The point of the number is narrower: a chemistry-key hold-out costs about one-fifth of the R^2 a topology hold-out costs, so the topology protocol is doing something beyond a chemistry hold-out, and what that something is remains unidentified.

The penalty is not specific to random forests. The two model families that reach comparable random-split accuracy agree on the topology penalty to within 0.0001 R^2 (0.3743 for random forest against 0.3742 for gradient boosting), and ridge shows a smaller absolute gap (0.180) that is largely a level difference — its random-split R^2 is 0.440 against 0.753. Normalising each family by its own random-split R^2 leaves ridge at 40.9% against 49.7% (random forest) and 51.1% (gradient boosting), so the normalisation does not bring the three together and no claim of model equivalence is made; these are single fits without tuning (SI-D). The penalty is also target-dependent — the same design on log-selectivity gives a smaller gap (0.8357 → 0.7211 against 0.7529 → 0.3786 for uptake) — so the magnitude quoted here is a property of the 0.15 bar, 298 K uptake target, not a universal constant of the protocol, and no single magnitude should be quoted without its target.

Leave-one-topology-out evaluation (a different protocol from the five-fold GroupKFold above) gives pooled R^2 = +0.3814 against weighted −0.0589, macro −0.5414 and median +0.2782 (Table 1). The two protocols are not interchangeable: the 5-fold GroupKFold trains five models on the folds of the 100,000-sample design, whereas LOTO trains one model per held-out topology (eleven models) on all other topologies and scores that topology alone, so their training pools and test folds are different objects. The sign of the aggregate depends on how folds are weighted: pooled scores every fold's prediction once against a single overall mean, while macro averages per-fold R^2 each normalised by its own variance, and a zero-variance fold makes R^2 undefined. Two different fold counts are involved and must not be merged: on the 100,000-sample single LOTO run there are eleven scoring folds, while the 25,000-sample multi-seed runs of SI-D use 10–11 valid folds per seed because a fold with zero target variance can drop out. Dropping the 5-structure rht fold moves the 100k macro from −0.5414 to −0.2948. The headline aggregation is chosen to match the intended deployment population, and the per-fold values are reported alongside; the positive pooled value is not produced by the pcu fold alone, as recomputing pooled while dropping only the pcu scoring fold (model and training folds unchanged) gives +0.4629. The per-fold values and the five-seed stability are in SI-D.

3.3 Fixed-test-set interventions reveal asymmetric partner effects

The pcu ↔ nbo contrast fixes the test set within each seed and changes only the training pool (Methods 2.4). MAE is the primary metric here because it is the deployment-relevant quantity — the average error in mmol/g that a screening model would carry — and it stays defined for the small single-topology test sets where R^2 is volatile; RMSE and the training-mean baseline are reported alongside, and the metric does not change the conclusion. The six arms are designed to answer one question in sequence — *is the degradation caused by how many training rows are removed, or by which ones?* Arm (a) is the full training pool (42,381 rows for nbo on average). Arm (b) removes the partner topology, leaving 10,656 rows on average. Arm (c) keeps the row count of (b) but draws the rows at random from (a), so it matches the size but not the identity of the removal. Arm (d) keeps the composition/topology profile of (b), which as shown below confines it to (b)'s own rows. Arm (e) removes only the partner rows whose composition recurs in the test set, and arm (f) is its size-matched draw. The comparisons are read as follows: (a)→(b) is the total effect of removing the partner; (a)→(c) asks whether random thinning to the same training size reproduces that effect; (b)→(c) asks which rows are retained at a fixed size; (e)→(f) asks the same size question inside arm (e)'s pool; and (b)→(f) is the size-matched version of the (e)→(b) comparison. No comparison isolates a single chemical mechanism.

On the nbo test set, removing the partner pcu raises MAE from 0.1065 ± 0.0032 to 0.1811 ± 0.0136 mmol/g — a paired difference of **+0.0746 mmol/g** (paired t-interval +0.0592 to +0.0900), which is **+70.0%** when scaled by the observed arm-(a) mean with that denominator treated as fixed — and RMSE rises by +0.1053 mmol/g (+77.9%, same scaling) (Figure 2, Table 2). The interval is a t-interval on the five absolute paired differences, not a ratio interval and not a per-seed percentage spread; its lower bound excludes zero, so the degradation is not within seed noise, and the interval is reported rather than a single point so the reader can see the effect's stability. A size-matched arm (c) that draws the same number of training rows at random from the full pool leaves MAE essentially unchanged (c: 0.1071 ± 0.0048, five outer seeds × five size-matched draws, averaged within seed before the seed-level SD; the seed-42 repeated-draw auxiliary c gives 0.1030 ± 0.0031, a distinct replication design kept in SI-C). The size-matched arm keeps 1,484 of the 1,532 compositions that arm (a) carries, so it preserves the chemical diversity while dropping the row count; its MAE is close to arm (a)'s (0.1071 against 0.1065), not to arm (b)'s (0.1811), so this random-thinning control does not reproduce the degradation. What differs between (b) and (c) is the identity and distribution of the retained rows at a fixed training size. The degradation therefore tracks **which rows** were removed,

not how many. Both arms still beat their respective training-mean baselines on the nbo test set, so the +70.0% is a genuine worsening of accuracy, not a re-ordering artefact (baseline errors and seed-level variability in SI-C).

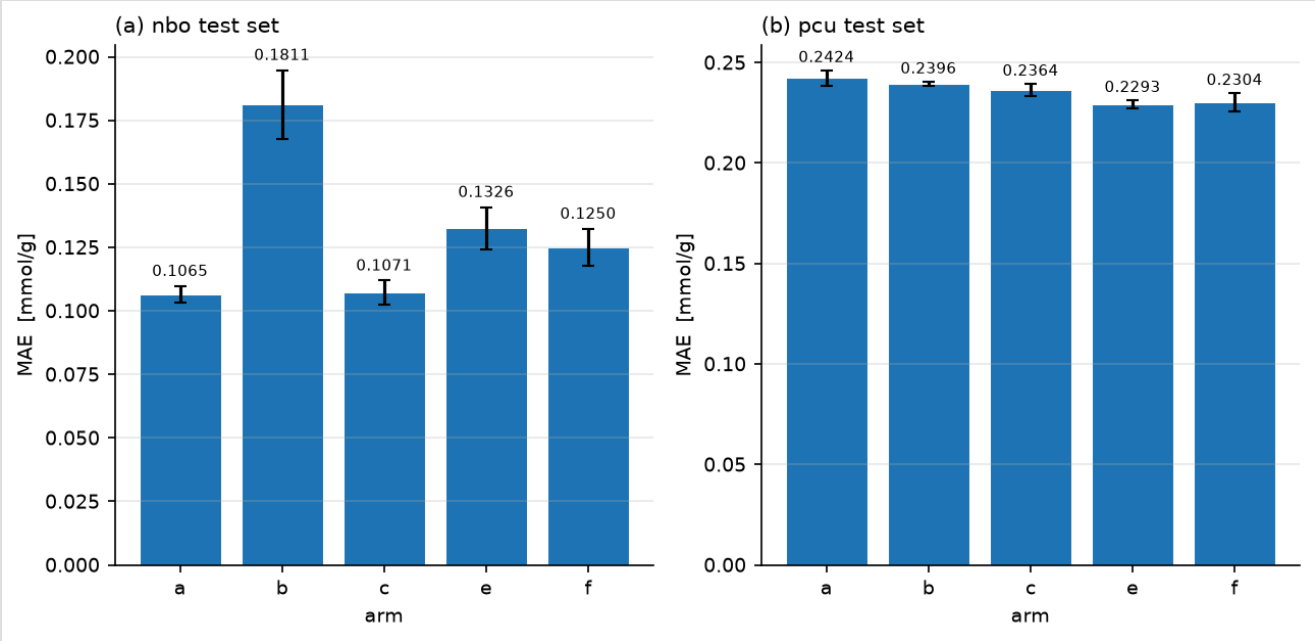


Figure 2. Fixed-test-set interventions on the pcu ↔ nbo pair (five outer seeds; test set fixed within each seed). Bars are five-seed means, error bars are ±1 SD across the five outer seed means (arm (c) is the nested 5 × 5 draw, its SD taken across seed means). Arm (d) is not plotted because it is confined to arm (b)'s own row set and therefore reproduces (b) by construction (SI-C.2). Left panel: nbo-side MAE per arm, where removing the partner (b) raises MAE by +70.0% while the size-matched draw (c) does not, and the shared-composition removal (e) and its size-matched draw (f) fall in between. Right panel: the same arms on the pcu side, where arm (e) improves MAE relative to (a), showing the effect is direction-dependent. *Source data:* `outputs/isoreticular_control_seeds.csv`, `outputs/composition_matched_control.csv`, `outputs/arm_e_partner_coverage.csv`.

The same intervention is not symmetric. On the pcu test set, removing the partner nbo leaves MAE essentially unchanged (−0.0028 mmol/g, paired t-interval −0.0080 to +0.0024, i.e. −1.2% of the arm-(a) mean) and RMSE essentially unchanged too (0.4030 ± 0.0021 to 0.4024 ± 0.0030, −0.1%), so the two sides of the pair behave differently from the outset: nbo's MAE degrades heavily when its pcu partner is removed, while the pcu-side change is small and its paired interval includes zero — the pcu side is far less sensitive to the partner, not independent of it. Both arms also beat their training-mean baselines on the pcu side; that comparison bounds them against a constant predictor, but beating a weak baseline does not exclude the model approaching its own error floor, so the near-zero change is read as no detected difference rather than as evidence of equivalence. This asymmetry foreshadows the direction dependence of the later arms. In deployment terms, the +70.0% is a large absolute worsening: an nbo-side screening model that carries 0.107 mmol/g error on average would carry 0.181 mmol/g if its pcu partner were unavailable. That increase could matter for screening, but the ranking effect of this intervention was not measured; §3.4 measures ranking under a separate 25,000-sample protocol contrast.

Two further arms qualify the result. Arm (e), which removes only those partner rows whose composition also occurs in the test set, raises nbo-side MAE to 0.1326 ± 0.0082 — more than the full pool but far less than removing the whole partner (0.1811 ± 0.0136) — while a size-matched draw (f) gives 0.1250 ± 0.0073, essentially unchanged from (e) and still far better than (b), confirming that the (e)→(b) step is not merely a size effect. The 34.9% / 65.1% split of the (a)→(b) degradation is an **ordering of magnitudes inside one contrast**, not a decomposition into two independent causal factors, and the two steps remove different coverage rather than a single shared quantity: arm (e) deletes the 12,342 partner rows that carry the 388 compositions shared with the test set (387 at the seed-42 baseline), leaving 1,144 of 1,532 compositions and 30,039 of 42,381 rows, while the further (e)→(b) step deletes the remaining

partner rows and loses a different 648 compositions (the remaining 65.1% of the degradation). Crucially, the effect is **direction-dependent**: on the pcu test set arm (e) improves MAE to 0.2293 against arm (a)'s 0.2424, and the pcu-side size-matched draw (f) gives 0.2304 ± 0.0045, still better than arm (b)'s 0.2396, so the direction dependence is not a size artefact. The paired pcu (e)−(a) difference is −0.0131 mmol/g (95% CI −0.0167 to −0.0096) and the (f)−(b) difference is −0.0092 mmol/g (−0.0144 to −0.0040); both intervals exclude zero, so the improvement is not a seed artefact. Post hoc, arm (g) removes the same number of partner rows at random, without conditioning on shared composition; it lies between (a) and (e) on both sides — pcu 0.2341 against a 0.2424 and e 0.2293; nbo 0.1189 against a 0.1065 and e 0.1326 — so shared composition is a consistent increment on top of a generic partner-removal effect (SI-C.4). Partner coverage is therefore not uniformly beneficial — a composition shared by two topologies carries two different mappings, and which one is removed matters. Arm (d), the composition-profile-matched draw, reproduces (b)'s composition/topology profile, which confines it to (b)'s own row set (five-seed means: 42,381 − 31,725 = 10,656; seed 42 gives 42,329 − 31,682 = 10,647, SI-C.2); its aggregated MAE is the same as (b)'s by construction (nbo 0.1811 ± 0.0136; pcu 0.2396 ± 0.0010), so it is a construction-consistency check, not a second independent mechanism contrast (SI-C).

The six arms establish three facts and refuse a fourth. Removing the partner is a large effect (a→b, +70.0% MAE, +77.9% RMSE); random thinning to the same size does not reproduce it (a→c); and the effect is direction-dependent (pcu-side e is better than a). The arms do not decompose the penalty into "coverage" and "diversity", because arms (e) and (f) move composition count and coverage together. This is one pair, five seeds, one model family, one target, so no single mechanism is identified; the direction dependence rules out the simplest "more data is always better" story and points to whether the removed rows carried the mapping the test side needed. The nbo-side degradation is positive in all five seeds, while the pcu-side (a)→(b) interval includes zero with the per-seed change reversing sign. Full per-seed rows are in SI-C.

3.4 Screening utility on a shared support set

Across five independently redrawn samples (25,000 rows; seeds 42, 143, 244, 345, 446), random-split validation recovered **50.4 ± 2.5%** of the true top 1% and topology-grouped validation **19.3 ± 7.1%** (mean ± SD), a fall of 31.1 percentage points (Figure 3). The top 1% is chosen as a representative screening budget: on a 25,000-material screen, 250 materials is the order of magnitude a synthesis-and-test budget would fund, so a metric computed at k = 1% tracks the deployment decision rather than a hypothetical full ranking. The intermediate strategies fill the ordering: chemistry key 44.0 ± 2.8%, linker pair 41.1 ± 4.1%, metal 29.2 ± 4.8%; this mirrors §3.2's R^2 order (chemistry key > linker pair > metal > topology), so the grouped-split penalty propagates from the fit into the deployable ranking rather than being an R^2-only artefact. The middle three strategies are reported because they interpolate the two endpoints: the metal-only hold-out (29.2%) sits far below the chemistry-key hold-out (44.0%), showing that holding out the coarsest chemical unit costs a large share of the true top-1% recovery. Recovery is far above the 1% chance level in every scheme — even the worst strategy selects ~19× the random expectation, against ~50× for random — but the gap between the best and worst protocol is the point, not the absolute level. Random performed best and topology worst in all five seeds; the complete five-strategy ordering held in four of five, so the two endpoint rankings are robust while the middle ordering is seed-sensitive and should not be over-read as a stable five-way hierarchy.

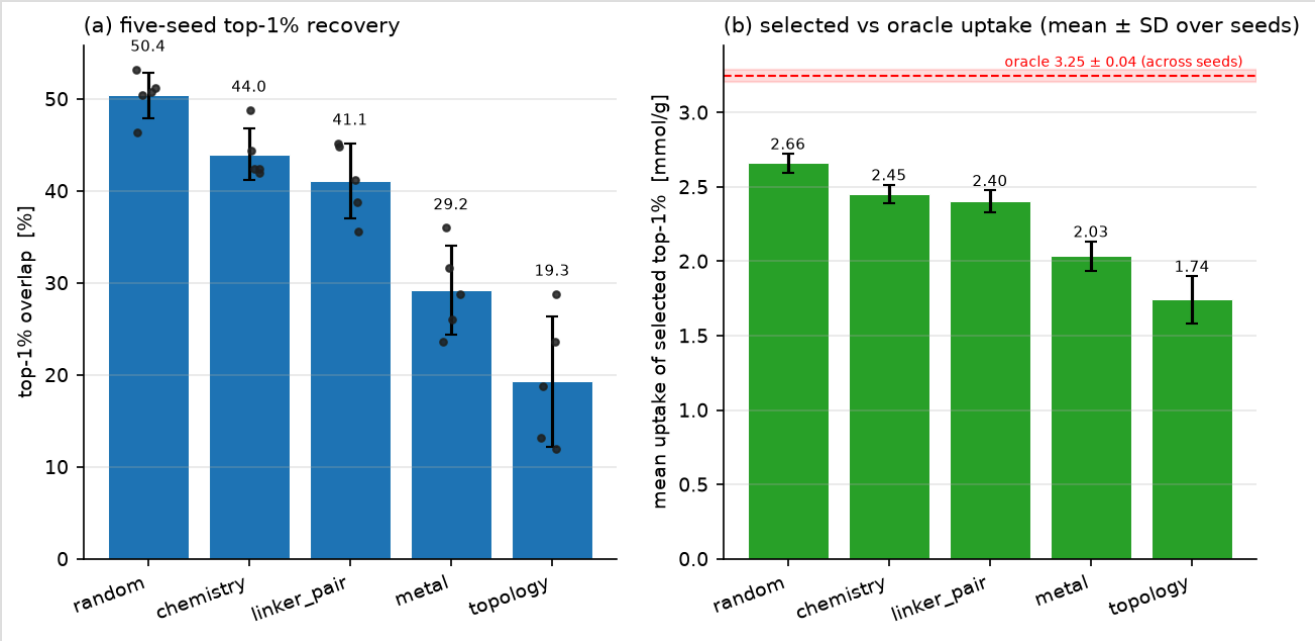


Figure 3. Five-seed screening utility on 25,000-sample support sets. **(a)** Top-1% overlap for the five strategies, bars are five-seed means, error bars ±1 SD, and the five filled dots are the per-seed values on the shared within-seed support set; random best and topology worst in all five seeds, the complete ordering holding in four. **(b)** Mean true uptake of the selected top-k, in the same arm order; the dashed line is the mean oracle uptake (≈3.25 mmol/g) from the same seeds. *Source data:* `outputs/screening_utility_seed{42,143,244,345,446}.csv`.

Materials selected under topology-grouped validation averaged 1.742 ± 0.158 mmol/g, **46.3% below** the mean oracle uptake (one minus the ratio of mean selected to mean oracle uptake; the per-seed mean of ratios rounds to the same 46.3% but is a distinct statistic). Overlap and regret are complementary: overlap counts how many of the true top-k were found, regret is the oracle uptake minus the selected uptake. At the seed-42 baseline the topology arm selects 1.587 mmol/g against an oracle 3.319 (regret 1.732), while random selects 2.706 (regret 0.613). Widening the budget to k = 5% and 10% attenuates but does not erase the gap (full table in SI-F).

The topology overlap ranges from 0.120 to 0.288 across seeds (sd 0.071, 37% of its mean) against random's far tighter 0.025, so a single run can fall either side of the typical value. The 31.1-percentage-point gap means topology-grouped validation hands over about three-fifths (61.7%) fewer true hits for the same budget, and this propagates from the fit into the ranked shortlist. These are contrasts on support sets shared within each seed; the random-split score is not an attainable baseline for deploying to genuinely unseen topologies, because no in-topology training data exist there.

3.5 A bounded support audit in ARC-MOF

The same support audit on ARC-MOF (264,225 structures, of which 256,351 carry both metal and linker cluster keys — the other 7,874 lack them and are excluded; topology and cluster metadata only; Figure 4a–c and Table 3) gives a coarse single-topology fraction of 99.3%, which remains **97.3%** when restricted to the 49,869 non-singleton compositions, and a fine non-singleton rate of **98.2%** (26,380 non-singletons) (Table 3). Restricting to non-singletons matters because a singleton composition is single-topology by construction; the high value therefore does not reduce to a sparse-count artefact, and the non-singleton figure is the more resistant claim. The near-total coupling is therefore not carried by compositions that appear once: it survives when those are excluded, and it survives again under the finer functional-group key, so the two robustness checks together close the two most obvious escape routes for the "it is just singletons" objection. The fine figure runs higher than the coarse one for the same reason it does in hMOF — the functional-group key fragments the composition space into more, smaller cells, each of which is more

likely to be single-topology — and it is reported next to the coarse value precisely so that the reader does not mistake the finer key for a cleaner separation. The same entropy-correction audit that demotes hMOF's fine figure to a robustness check is applied here: the conditional-entropy correction is (K_TC − K_C)/(2 N ln 2), with occupied marginal/joint cell counts of 180,989/184,978 (coarse) and 177,971/178,834 (fine), giving corrections of 0.0112 bit (15.2%) and 0.0029 bit (19.8%) against H(T|C) = 0.0739 and 0.0147 bit, so both ARC-MOF values survive the correction. The audit consumes only topology and cluster metadata, so it is a support audit, not an external prediction test on adsorption properties ARC-MOF does not carry in the form this study's models consume.

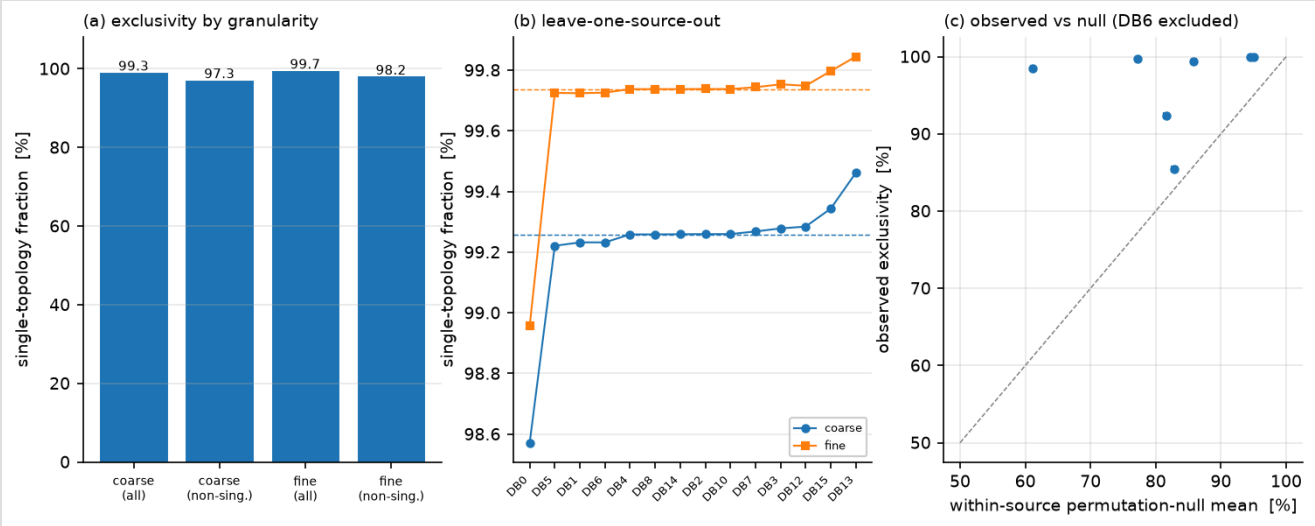


Figure 4. Bounded support audit in ARC-MOF. **(a)** Single-topology composition fraction at four combinations (coarse/fine × all/non-singleton): 99.3, 97.3, 99.7 and 98.2. **(b)** Leave-one-source-out sensitivity: removing DB0 moves the fraction to 98.57–98.96, any other single source to 99.22–99.84. **(c)** Observed versus within-source topology-permutation null on the common support set, with the degenerate single-topology source (DB6) excluded; the permutation SD is below 0.5%, so the horizontal error bars are not visually distinguishable. *Source data:* `outputs/confound_arcmof.csv`, `outputs/confound_arcmof_excl_derived.csv`, `outputs/common_support/null_topology_cardinality.csv`. ---

ARC-MOF is **not independent of hMOF**: it describes itself as spanning public MOF databases and shares authors with hMOF, so a zero file-name intersection proves nothing about content overlap. The source-stratified audit is possible at all because ARC-MOF ships its own provenance as file-name prefixes (DB0 … DB15), a metadata affordance that enumerated libraries should provide but which is not guaranteed elsewhere. Removing one source sub-library at a time (the DB0 condition stated rather than assumed away) bounds the effect of any overlap: removing the largest sub-library DB0 (201,093 rows) moves the single-topology fraction to 98.57% (coarse) and 98.96% (fine), and removing any single other sub-library leaves it between 99.22% and 99.46% at the coarse granularity and between 99.72% and 99.84% at the fine granularity, at the cost of discarding 79.0% of the analysed rows (169,489 of 214,543 at the fine granularity) — a stress test on a fifth of the data, not a worst case over all of it, and the bound holds only if the hMOF-derived rows are concentrated in DB0. Within each source, the observed single-topology fraction sits above its own topology-permutation null on a common support set, with excesses from +2.6 (DB15) to +37.3 (DB12) percentage points (for example, DB0 observes 99.7% against a null of 77.2%, DB12 observes 98.4% against 61.1%, DB15 observes 85.4% against 82.8%), and the degenerate single-topology source excluded. The excess shows a moderate-to-strong positive point estimate against the source's topology entropy (Pearson r = +0.67, p = 0.10), but there are only seven non-degenerate sources, so this is weak evidence either way; removing the single influential source DB12 flips the point estimate (Pearson r = −0.08, Spearman ρ −0.09), so the association is not established. The source-to-source spread itself is informative: at the common-support coarse granularity the exclusivity rate varies from 85.4% to 100.0% (86.1% to 100.0% at the fine granularity), so the audit reveals source-associated heterogeneity rather than a single number shared by all enumerated libraries, and that

heterogeneity is why the per-source audit, not a single database-wide value, carries the ARC-MOF claim. The audit does not identify which construction choices caused that heterogeneity.

Two controls prevent the audit from being read as stronger than it is, and they address different null hypotheses. The leave-one-source-out range (98.6–99.8%) is a partial, not a complete, bound on cross-source overlap, for the DB0 condition stated above. The **within-source control** permutes topology labels across records within each source, holding composition keys and label counts fixed (1,000 permutations), and rejects "the observed association is what a within-source label shuffle would produce". Separately, a **common-rate binomial parametric bootstrap** (20,000 draws) tests whether the between-source spread exceeds a single-common-rate sampling model. Neither control identifies a database-construction mechanism. The per-source exclusivity rates also carry a bootstrap spread (10 resamples per source, reported in SI-H), so between-source differences are not read off single point estimates. Because the two databases define "a composition" with different keys and cover different row sets, no ordering of their magnitudes is claimed; this is a bounded support audit under a second set of metadata definitions, not an independent replication.

Under ARC-MOF's own, non-equivalent composition definitions, the coupling is again near-total and survives the non-singleton restriction and the permutation control — a bounded yes to "does the coupling reappear under a different keying?", not an independent replication.

---

4. Discussion

4.1 What this means for the stress-test protocol

Quantifying the removed support changes how a topology hold-out is read, not whether it should be used. It remains a legitimate deployment proxy for structural extrapolation; what the audit rules out is reading it as an isolated test of topological transfer. Three quantities must be kept separate, and the distinction can be made concrete. The **identity** of what a split removes is a structural fact established by counting, with no model: 74.7% of coarse compositions are private to one topology. The **protocol contrast** is a ratio between two validation schemes re-run on a shared support set: the chemistry-key decrease is 21.1% of the topology decrease. The **consequence** is a local intervention result on one pair: removing the partner topology raised nbo-side MAE by 70.0%, while removing only the shared-composition partner rows improved pcu-side MAE — two different interventions, not one operation that reverses sign. Conflating the first with the third is what lets a correct observation — 74.7% of compositions are single-topology — be read as "74.7% of the penalty is chemistry", which the data do not establish. The two percentages answer different questions and must not be joined: the support audit's 74.7% measures *what* the split withholds (the chemistry private to a topology), whereas §3.2's 21.1% is the ratio of the chemistry-key decrease to the topology decrease; the former is a count, the latter is a ratio between two validation protocols, and one cannot be read off the other — a split can withhold 74.7% of the chemistry and yet cost only 21.1% of the R^2 in a chemistry-key comparison.

The protocol is therefore better described as a combined structural-and-chemical stress test than as a test of topology alone. That description does not make it useless, only more honestly scoped: a hold-out that withholds both a structural family and the chemistry attached to it is still a harder test than random splitting, and its score still carries information about deployment to genuinely novel structures. What it does not carry is a licence to attribute the score to topology per se. We do not advocate abandoning topology hold-out; we advocate reporting its chemical support alongside its score. The correction is cheap — the support audit is a few seconds of counting on data the reporter already holds — and it is additive: a report that already states its folds and its metric need only add the per-group

unseen fraction and the overlap between its grouping key and its chemistry key, which is one added sentence rather than a new analysis.

4.2 Relation to prior work and mechanistic hypotheses

This work extends, rather than corrects, the 2026 studies that already treat topology hold-out as a combined stress test — Asgari et al.'s split-only experiment and Alimardani et al.'s multi-target, multi-descriptor sweep both reach that framing before this paper, and neither is contradicted here. Its addition is the measured support loss and a usable way to make that loss visible before training, turning the qualitative "combined stress test" into a quantitative "here is what the split withholds". On the mechanism, the omitted-variable analysis is consistent with linker chemistry being the identity information the scalar descriptors capture least: group-associated η^2 fell from 0.168 to 0.031 for topology and from 0.393 to 0.255 for the linker pair after geometric-model residualisation, and linker-pair labels predicted the residual with $R^2 = 0.214$. These are descriptive residual associations, not a decomposition of the topology-hold-out penalty; the two η^2 values have different total-variance denominators, so their ratio is not "the share of variance absorbed". The identity block is also only useful where identity information crosses the fold boundary — across the three grouping variables, the within-split identity effect rises with the availability of the non-grouped identity information, from −0.011 under topology grouping (where the group label itself is held out) to +0.032 under the grouping that leaves the most identity information unremoved — so its effect under a grouped split is a construction consequence, not a sufficient mechanism; the availability ordering is reported as a regularity to be explained, not as an explanation itself. Both of these are predictive associations under a single model and feature set, not demonstrations of a mechanism, and the controlled contrast of §3.3 does not separate chemical coverage, composition richness and descriptor support. The ten-target panel is descriptive: the split effect is positive on 10/10 targets (median +0.324, Wilcoxon $p = 0.0020$), and the within-split identity effect is not distinguishable from zero for topology and metal, but its targets are algebraically dependent, so $n = 10$ is not ten independent units and its p-values are not confirmatory. These supporting analyses are in SI-E and SI-G; the main text keeps only the conclusion that the mechanism remains unidentified, because every analysis here is "consistent with" rather than "demonstrated", and the latter would require an orthogonal intervention the design does not provide.

4.3 Three reporting principles

Three actions make a grouped-split claim testable, and each corresponds to a quantity this paper had to report before its own claims became readable. **First**, report the overlap between the grouping variable and the chemistry key, and the per-group unseen fraction, next to every grouped score — this is the quantity a reader needs to know whether "topology hold-out" is partly a chemistry hold-out, and it costs only a few seconds of counting on data the reporter already holds; here it is the 74.7% / 7-of-11 statement that precedes any R^2. A report that omits it forces the reader to guess, and a reader who guesses "none" will over-read the score. **Second**, fix the comparison support set and report the aggregation and ranking metric that match the deployment question; a single pooled R^2 does not disclose the sign and volatility of the per-group view, so both the aggregate and the per-fold values must be given, and the headline aggregation should follow the deployment population rather than a fixed rule — here the pooled +0.3814 and macro −0.5414 are reported together because neither alone describes the outcome. A report that gives only pooled when the groups are imbalanced hides the sign reversal that macro reveals. **Third**, state the effective number of groups, the replication design, the chemistry granularity and any source dependence, because each of these moves the reported magnitude and none can be inferred by a reader — here the five seeds, the two size-matched designs, the coarse/fine distinction and the ARC-MOF source dependence are all stated in the results, not left to the Supporting Information. The three principles are ordered by cost: the first is a count, the second a reporting choice, the third a disclosure, and

none requires new modelling. The complete 22-rule checklist and its companion diagnostic are in the Supporting Information (SI-A) and the released tooling; these three principles are its condensed form, not a replacement for it.

4.4 Limitations and testable designs

Four limits bound the claim, and each is stated next to the result it constrains rather than buried in the Supporting Information. (i) The audit is established in hMOF (an enumerated library) and in ARC-MOF (a mixed-source library that is partly dependent on hMOF); extending it to experimentally derived libraries — curated collections such as CoRE MOF (ref. 5) are the obvious first targets — requires running the same cheap diagnostic there, not assuming transfer, because experimental libraries are not built by recombining a finite block set on templates and may not show the same composition-topology coupling. (ii) Only one chemistry-matched pair supports a controlled intervention, so the directional effect is demonstrated on a single pair, not across a population of topologies; a single pair is not a population, and the direction dependence could be specific to pcu and nbo. (iii) The intervention cannot vary chemical coverage, composition richness and descriptor support independently — they move together when a partner topology is removed — so the contrast excludes training-set size but does not isolate a unique mechanism. (iv) Generalisation beyond the model family, the target and the sample scales used here is untested: beyond random forest, gradient boosting and ridge, beyond uptake and log-selectivity, and beyond the 25,000–100,000 sample sizes, no claim is made, and each of these is an untested axis rather than an assumed one. These four limits bound the three results in the terms of §4.1 — a structural fact, a ratio between two protocols, a one-pair intervention — and they do not support the convenient stronger reading, "the topology penalty is mostly a chemistry penalty".

A further alternative interpretation of the direction dependence should be stated rather than left implicit. pcu is roughly four times larger than nbo in the full database (205,361 against 49,464 structures) and chemically broader, so removing the partner in arm (e) removes 12,342 training rows on the nbo side but only 6,689 on the pcu side. Arm (c) excludes the row count *within* each side — it draws the same number of rows while keeping 1,484 of 1,532 compositions — but it does not exclude the cross-side asymmetry that pcu is inherently larger and more diverse than nbo. The direction dependence is therefore consistent with a size/diversity asymmetry rather than with a general, side-independent property of partner coverage, and this alternative is listed as a testable hypothesis rather than resolved here; a second chemistry-matched pair would be the direct way to adjudicate it, but hMOF offers no second usable pair, so the adjudication depends on a library that supplies one.

The actionable next step is not "more data" for the same pair; it is a targeted extension of the single usable contrast. Concretely, one would identify several chemistry-matched topology pairs beyond pcu ↔ nbo — a step the overlap diagnostic makes cheap, but which hMOF itself cannot supply, because the current library offers only the one usable pair, so the additional pairs must come from another library — fix the test distribution within each seed exactly as here, and re-run arms (a), (b), (c), (e) and (f) on each pair, so that the direction dependence and the size-match control can be tested across a population of pairs rather than a single one. Running the same support diagnostic on experimental libraries whose construction is not enumerative would then say whether the coupling is an artefact of enumeration or a property of MOF chemistry more generally. The two extensions are cheap, additive and pre-registerable, and neither needs a stronger model or more rows of the same data.

---

5. Conclusion

Topology hold-out removes chemical support as well as topology labels in hMOF, with 74.7% of coarse compositions private to one topology and 61.3% of structures losing their same-key counterpart; a second, partly dependent database (ARC-MOF) shows the same coupling under bounded sensitivity. Quantifying this loss clarifies

the validation task but does not identify the causes of the prediction penalty. Fixed-test-set interventions on the usable pcu ↔ nbo pair revealed asymmetric, direction-dependent effects not explained by random thinning to the same size, and five-seed screening showed a substantial reduction in top-ranked recovery under topology-grouped validation. A support audit in ARC-MOF, bounded by source dependence and singleton controls, confirms strong coupling under a second set of metadata definitions. We retain topology hold-out as a stress test while recommending that reports disclose chemical support, deployment-relevant aggregation and ranking performance, so that the scope of a grouped-split claim is testable without presenting it as a clean separation of topological and chemical transfer.

---

7. Tables

Table 1. Validation protocols, two panels. **Panel A:** five schemes on 100,000 samples, pooled R^2 under 426 features and the matched 415-feature check. **Panel B:** leave-one-topology-out aggregates of the same 100,000-sample run; the 25,000-sample five-seed LOTO replication is Table S9.

Panel A — five schemes (pooled R^2).

Scheme	426 features	415 features (topology block removed)
random	0.7529	0.7449
chemistry key	0.6738	0.6674
linker pair	0.6606	—
metal	0.5270	—
topology (GroupKFold)	0.3786	0.3845

Panel B — leave-one-topology-out aggregates.

Aggregation	pooled	weighted	macro	median
LOTO R^2	+0.3814	−0.0589	−0.5414	+0.2782

Source data: `outputs/leakage_results_final.csv`, `outputs/leakage_results_chemistry.csv`, `outputs/leakage_results_notopo100k.csv`, `outputs/metrics_audit.csv`.

Table 2. Controlled-contrast MAE (mean ± SD over five outer seeds), nbo and pcu test sets. n_train, MAE and RMSE are all five-seed means; the (c) SD is the nested seed-level value.

Test	arm	n_train	MAE	RMSE
nbo	a (with partner)	42,381	0.1065 ± 0.0032	0.1352 ± 0.0077
nbo	b (partner removed)	10,656	0.1811 ± 0.0136	0.2405 ± 0.0136
nbo	c (size-matched)	10,656	0.1071 ± 0.0048	0.1375 ± 0.0109
nbo	e (shared-composition removed)	30,039	0.1326 ± 0.0082	0.1745 ± 0.0127
nbo	f (size-matched from e)	10,656	0.1250 ± 0.0073	0.1697 ± 0.0145
pcu	a (with partner)	18,275	0.2424 ± 0.0038	0.4030 ± 0.0021
pcu	b (partner removed)	10,656	0.2396 ± 0.0010	0.4024 ± 0.0030
pcu	c (size-matched)	10,656	0.2364 ± 0.0031	0.4002 ± 0.0044

Test	arm	n_train	MAE	RMSE
pcu	e (shared-composition removed)	11,586	0.2293 ± 0.0020	0.3935 ± 0.0032
pcu	f (size-matched from e)	10,656	0.2304 ± 0.0045	0.3961 ± 0.0066

Note: The (c) arm is the nested size-matched draw (`diversity_control.py`; five outer seeds × five draws, SD across seed means). The seed-42 repeated-draw (c) is a distinct replication design in SI-C. *Source data:* `outputs/isoreticular_control_seeds.csv`, `outputs/composition_matched_control.csv`, `outputs/arm_e_partner_coverage.csv`.

Table 3. ARC-MOF support audit under its own chemistry definitions.

granularity	N	M	non-singleton	single-topology (all)	single-topology (non-singleton)
coarse	256,351	180,989	49,869	99.3%	97.3%
fine	214,543	177,971	26,380	99.7%	98.2%

Source data: `outputs/confound_arcmof.csv` (N, M and all-composition fractions; `topo_col = Crystalnet`) and `outputs/arcmof_non_singleton_summary.csv` (non-singleton counts and fractions); the singleton counts and entropy corrections behind §3.1 and §3.5 are in `outputs/singleton_audit.csv`. Source-removal and null tables are in SI-H.

---

8. References

1. Wilmer, C. E.; Leaf, M.; Lee, C. Y.; Farha, O. K.; Hauser, B. G.; Hupp, J. T.; Snurr, R. Q. Large-scale screening of hypothetical metal–organic frameworks. *Nat. Chem.* **2012**, *4*, 83–89. DOI: 10.1038/nchem.1192
2. Boyd, P. G.; Chidambaram, A.; García-Díez, E.; Ireland, C. P.; Daff, T. D.; Bounds, R.; Gladysiak, A.; Schouwink, P.; Moosavi, S. M.; Maroto-Valer, M. M.; Reimer, J. A.; Navarro, J. A. R.; Woo, T. K.; Garcia, S.; Stylianou, K. C.; Smit, B. Data-driven design of metal–organic frameworks for wet flue gas CO_2 capture. *Nature* **2019**, *576*, 253–256. DOI: 10.1038/s41586-019-1798-7
3. Burner, J.; Luo, J.; White, A.; Mirmiran, A.; Kwon, O.; Boyd, P. G.; Maley, S.; Gibaldi, M.; Simrod, S.; Ogden, V.; Woo, T. K. ARC–MOF: A Diverse Database of Metal-Organic Frameworks with DFT-Derived Partial Atomic Charges and Descriptors for Machine Learning. *Chem. Mater.* **2023**, *35* (3), 900–916. DOI: 10.1021/acs.chemmater.2c02485
4. Bobbitt, N. S.; Shi, K.; Bucior, B. J.; Chen, H.; Tracy-Amoroso, N.; Li, Z.; Sun, Y.; Merlin, J. H.; Siepmann, J. I.; Siderius, D. W.; Snurr, R. Q. MOFX-DB: An Online Database of Computational Adsorption Data for Nanoporous Materials. *J. Chem. Eng. Data* **2023**, *68* (2), 483–498. DOI: 10.1021/acs.jced.2c00583
5. Chung, Y. G.; Haldoupis, E.; Bucior, B. J.; Haranczyk, M.; Lee, S.; Zhang, H.; Vogiatzis, K. D.; Milisavljevic, M.; Ling, S.; Camp, J. S.; Slater, B.; Siepmann, J. I.; Sholl, D. S.; Snurr, R. Q. Advances, Updates, and Analytics for the Computation-Ready, Experimental Metal–Organic Framework Database: CoRE MOF 2019. *J. Chem. Eng. Data* **2019**, *64* (12), 5985–5998. DOI: 10.1021/acs.jced.9b00835
6. Jablonka, K. M.; Ongari, D.; Moosavi, S. M.; Smit, B. Big-Data Science in Porous Materials: Materials Genomics and Machine Learning. *Chem. Rev.* **2020**, *120* (16), 8066–8129. DOI: 10.1021/acs.chemrev.0c00004
7. Moosavi, S. M.; Nandy, A.; Jablonka, K. M.; Ongari, D.; Janet, J. P.; Boyd, P. G.; Lee, Y.; Smit, B.; Kulik, H. J. Understanding the diversity of the metal-organic framework ecosystem. *Nat. Commun.* **2020**, *11*, 4068. DOI: 10.1038/s41467-020-17755-8

8. Alimardani, H.; Abaei, S.; Asgari, M. Descriptor Adequacy as a Materials-Regime Principle in MOF Adsorption Machine Learning. *ChemRxiv* **2026**, version 2 (2026-07-03). DOI: 10.26434/chemrxiv.15002252/v2
9. Abaei, S.; Masoudi, F.; Alimardani, H.; Asgari, M. Mapping trust in MOF adsorption predictions: chemically auditable descriptor atlases for machine-learning screening. *J. Mater. Chem. A* **2026**, *14*, 41247–41262. DOI: 10.1039/D6TA05505B
10. Abaei, S.; Alimardani, H.; Fakhroleslam, M.; Eslami, F.; Asgari, M. Benchmark maturity in MOF adsorption machine learning: when do conclusions become scientifically reliable? *Digital Discovery* **2026**, *5*, 4541–4553. DOI: 10.1039/D6DD00351F
11. Pedregosa, F.; Varoquaux, G.; Gramfort, A.; Michel, V.; Thirion, B.; Grisel, O.; Blondel, M.; Prettenhofer, P.; Weiss, R.; Dubourg, V.; Vanderplas, J.; Passos, A.; Cournapeau, D.; Brucher, M.; Perrot, M.; Duchesnay, É. Scikit-learn: Machine Learning in Python. *J. Mach. Learn. Res.* **2011**, *12* (85), 2825–2830. URL: <https://jmlr.org/papers/v12/pedregosa11a.html>
12. Harris, C. R.; Millman, K. J.; van der Walt, S. J.; Gommers, R.; Virtanen, P.; Cournapeau, D.; Wieser, E.; Taylor, J.; Berg, S.; Smith, N. J.; et al. Array programming with NumPy. *Nature* **2020**, *585*, 357–362. DOI: 10.1038/s41586-020-2649-2
13. McKinney, W. Data Structures for Statistical Computing in Python. *Proceedings of the 9th Python in Science Conference* **2010**, 56–61. DOI: 10.25080/majora-92bf1922-00a
14. Hunter, J. D. Matplotlib: A 2D Graphics Environment. *Comput. Sci. Eng.* **2007**, *9* (3), 90–95. DOI: 10.1109/MCSE.2007.55
15. Burner, J.; Luo, J.; White, A.; Mirmiran, A.; Kwon, O.; Boyd, P. G.; et al. *ab initio* REPEAT Charge MOF Database (ARC-MOF), version 6 [data set]. Zenodo, 2024. DOI: 10.5281/zenodo.13891643
16. Asgari, M.; Abaei, S.; Di Bella, A.; Hadji-Thomas, A.; Alimardani, H. The Validation Split Defines the Discovery Claim in AI-Guided MOF Screening. *ChemRxiv* **2026**, version 1 (2026-06-15). DOI: 10.26434/chemrxiv.15004762/v1
17. Wang, R.; Gao, Y.; Kang, L.; Wu, Y.; Zhu, L. A systematic assessment of data splitting strategies for machine learning prediction of pressure-specific CO_2 uptake in hypothetical metal–organic frameworks (hMOFs). *Comput. Theor. Chem.* **2026**, *1264*, 115959. DOI: 10.1016/j.comptc.2026.115959
18. Roberts, D. R.; Bahn, V.; Ciuti, S.; Boyce, M. S.; Elith, J.; Guillera-Arroita, G.; Hauenstein, S.; Lahoz-Monfort, J. J.; Schröder, B.; Thuiller, W.; Warton, D. I.; Wintle, B. A.; Hartig, F.; Dormann, C. F. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography* **2017**, *40*, 913–929. DOI: 10.1111/ecog.02881
19. Valavi, R.; Elith, J.; Lahoz-Monfort, J. J.; Guillera-Arroita, G. blockCV: An R package for generating spatially or environmentally separated folds for k-fold cross-validation of species distribution models. *Methods Ecol. Evol.* **2019**, *10*, 225–232. DOI: 10.1111/2041-210X.13107
20. Kapoor, S.; Narayanan, A. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns* **2023**, *4* (9), 100804. DOI: 10.1016/j.patter.2023.100804
21. Wu, Z.; Ramsundar, B.; Feinberg, E. N.; Gomes, J.; Geniesse, C.; Pappu, A. S.; Leswing, K.; Pande, V. MoleculeNet: a benchmark for molecular machine learning. *Chem. Sci.* **2018**, *9*, 513–530. DOI: 10.1039/C7SC02664A
22. Yang, K.; Swanson, K.; Jin, W.; Coley, C.; Eiden, P.; Gao, H.; Guzman-Perez, A.; Hopper, T.; Kelley, B.; Mathea, M.; Palmer, A.; Settels, V.; Jaakkola, T.; Jensen, K.; Barzilay, R. Analyzing learned molecular representations for property prediction. *J. Chem. Inf. Model.* **2019**, *59*, 3370–3388. DOI: 10.1021/acs.jcim.9b00237

23. Sheridan, R. P. Time-split cross-validation as a method for estimating the goodness of prospective prediction. *J. Chem. Inf. Model.* **2013**, *53*, 783–790. DOI: 10.1021/ci400084k
24. Xie, T.; Grossman, J. C. Crystal Graph Convolutional Neural Networks for an Accurate and Interpretable Prediction of Material Properties. *Phys. Rev. Lett.* **2018**, *120*, 145301. DOI: 10.1103/PhysRevLett.120.145301
25. Lundberg, S. M.; Lee, S.-I. A Unified Approach to Interpreting Model Predictions. *Advances in Neural Information Processing Systems* **2017**, *30*, 4765–4774. URL: <https://proceedings.neurips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html>
26. Rosen, A. S.; Iyer, S. M.; Ray, D.; Yao, Z.; Aspuru-Guzik, A.; Gagliardi, L.; Notestein, J. M.; Snurr, R. Q. Machine learning the quantum-chemical properties of metal–organic frameworks for accelerated materials discovery. *Matter* **2021**, *4* (5), 1578–1597. DOI: 10.1016/j.matt.2021.02.015
27. Bucior, B. J.; Rosen, A. S.; Haranczyk, M.; Yao, Z.; Ziebel, M. E.; Farha, O. K.; Hupp, J. T.; Siepmann, J. I.; Aspuru-Guzik, A.; Snurr, R. Q. Identification Schemes for Metal–Organic Frameworks To Enable Rapid Search and Cheminformatics Analysis. *Cryst. Growth Des.* **2019**, *19* (11), 6682–6697. DOI: 10.1021/acs.cgd.9b01050
28. Abaei, S.; Alimardani, H.; Masoudi, F.; Aidi, M.; Asgari, M. When does framework topology matter beyond pore metrics in MOF adsorption prediction? *Sep. Purif. Technol.* **2026**, *414*, 139617. DOI: 10.1016/j.seppur.2026.139617
29. Tossou, P.; Wognum, C.; Craig, M.; Mary, H.; Noutahi, E. Real-World Molecular Out-Of-Distribution: Specification and Investigation. *J. Chem. Inf. Model.* **2024**, *64* (3), 697–711. DOI: 10.1021/acs.jcim.3c01774
30. Li, S.; Sun, P. Toward More Trustworthy QSAR: A Systematic Discussion on Data Set Partitioning. *J. Chem. Inf. Model.* **2026**, *66* (4), 2199–2210. DOI: 10.1021/acs.jcim.5c02465
31. Almendra, F.; Falcão, A. O. Validation Strategy Dominates Model Choice in Chronological QSAR Evaluation. *J. Chem. Inf. Model.* **2026**. DOI: 10.1021/acs.jcim.6c02796
32. Friedman, J. H. Greedy function approximation: a gradient boosting machine. *Ann. Stat.* **2001**, *29* (5), 1189–1232. DOI: 10.1214/aos/1013203451
33. Hoerl, A. E.; Kennard, R. W. Ridge regression: biased estimation for nonorthogonal problems. *Technometrics* **1970**, *12* (1), 55–67. DOI: 10.1080/00401706.1970.10488634

---

9. Declarations

Data availability

Code and data (full links). Code and derived results: <https://github.com/wang359826/topology-holdout-support-audit> (tag `v1.3`), archived at <https://doi.org/10.5281/zenodo.23434268> (version DOI, the version used for this manuscript); the concept DOI <https://doi.org/10.5281/zenodo.23415179> resolves to the latest version. This repository was created under an earlier working title that read the topology penalty as "largely" a chemistry penalty; that wording predates the findings reported here and is not the claim of this manuscript, which reports the support loss (74.7% of coarse compositions are single-topology) and a descriptive protocol ratio (the chemistry-key hold-out costs 21.1% of the topology hold-out's R^2) and does not attribute the topology penalty to chemistry. The earlier title is retained on the preceding Zenodo record only for link stability. Input data: hMOF at <https://archive.materialscloud.org/records/nsp7v-2dk91> and ARC-MOF v6 at <https://doi.org/10.5281/zenodo.13891643>.

This work generates **no new experimental or simulation data**. **Neither input database is redistributed with this manuscript or with the source-code archive**; both are publicly available without registration and are fetched by the reader. The source-code archive's `DATA.md` gives the exact retrieval commands, the expected byte sizes and the MD5 of every file used, and records two known upstream failure modes so a reader is not misled by them. Licences are those stated on the two record pages. Archived analysis outputs and final figures are stored at the locations specified in `REPRODUCE.md`; historical and newly generated outputs have separately documented verification and checksum scopes, and the Supporting Information and `REPRODUCE.md` state what was verified end to end and what was not.

Code availability

Analysis scripts, environment specifications and data-retrieval instructions are provided in the accompanying source-code archive. The reproduction supplement contains the archived outputs, the final-figure and PDF builders, and the standalone training-mean baseline generator; `REPRODUCE.md` identifies the required package and entry point for each reported result. Dependencies are five standard Python packages for levels 1–4; the optional level-5 SHAP attribution additionally uses `shap`, and **no GPU or deep-learning framework is required**. No level-1–4 timing was taken on the step list shipped today: levels 2 and 3 were timed on 6- and 8-step lists, level 1 before `full_db_confound.py` joined, and level 4 on a 6-step list since changed; the current lists are 7 / 10 / 12 / 8 steps, and every step added since its level's timing has been re-run separately with byte-identical outputs. Only level 5 was run on its current 7-step list, on 2026-10-14, with per-step logs in `outputs/logs/L01` – `L07`; the exact step lists, per-step timings and file-by-file comparison are in the source-code archive's `REPRODUCE.md`, §3. Re-running levels 2 and 3 on newer library versions reproduces the reported numbers to $\leq 10^{-6}$, byte-identity being a property of the original environment rather than of the code.

Author contributions (CRediT)

The sole author performed all roles: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing.

Conflicts of interest

There are no conflicts of interest to declare.

Funding

This work received no external funding.

Acknowledgements

This work was carried out **independently by the author**, as an undergraduate in the Chemical Engineering and Technology programme at Nanjing Tech University, with no external funding. All computations ran on a single laptop CPU. The author thanks the original authors of the two datasets, released publicly, on whose work this study depends and against whose conclusions it is checked.

AI usage disclosure. Large language models were used as **pair-programming and language-editing aids**. All experimental designs, variable definitions, statistical choices and conclusions were made by the author. AI-assisted text was reviewed and edited by the author before inclusion; numerical claims are traceable to the archived outputs and producer scripts named in `REPRODUCE.md`. The author takes full responsibility for the content.